Ramesh Singh

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Summary

Data Science enthusiast and B.Tech CSE student with hands-on experience in machine learning, deep learning, and generative AI. Proficient in building end-to-end ML pipelines, RAG-based LLM applications, and real-time computer vision systems using Python and modern AI frameworks. Experienced through AICTE-backed internships with Microsoft Azure and IBM SkillsBuild, with a strong foundation in model development, MLOps, and data-driven problem solving.

Education

Veer Bahadur Singh Purvanchal University — Jaunpur, UP
B.Tech in Computer Science and Engineering — *2022 – Present*

Naval's National Academy — Gorakhpur, UP
Intermediate – Class XII (PCM) — *2021 – 2022*
High School – Class X — *2019 – 2020*

Technical Skills

Languages & Web: Python, SQL, HTML, CSS
ML & Data Science: Scikit-learn, XGBoost, Random Forest, SVM, Regression, Clustering, PCA, NumPy, Pandas
Deep Learning & NLP: TensorFlow, Keras, CNN, LSTM, Transformers, NLTK, Sentiment Analysis
Generative AI: LangChain, FAISS, Google Gemini API, RAG
MLOps & Tools: Docker, MLflow, GitHub Actions, AWS, Git, VS Code
Visualization: Matplotlib, Seaborn, Streamlit, Flask

Projects

Chat with PDF using Gemini | *Python, Streamlit, LangChain, FAISS, Google Gemini API* | GitHub

- Built an end-to-end RAG pipeline for multi-PDF Q&A using Google Gemini and LangChain with FAISS-based semantic vector search for accurate context retrieval
- Deployed an interactive Streamlit web app with tuned chunk size/overlap for improved answer grounding and secure API key management

Campus Safety & Surveillance System | *Python, YOLOv8, OpenCV, DeepFace, Streamlit* | GitHub

- Developed a real-time AI surveillance system integrating YOLOv8 object detection, DeepFace recognition, and pose estimation with multi-threaded video stream processing
- Built a Streamlit dashboard for live monitoring and alert generation; optimized inference pipeline for CPU-only, resource-constrained environments

Internships

AICTE – Microsoft AI & Azure Virtual Internship — June 2025 (4 Weeks)
Edunet Foundation

- Deployed a cloud-based ML model using Azure AI services achieving 91% prediction accuracy; completed end-to-end solution within a 4-week sprint under industry mentorship

AICTE – IBM SkillsBuild AIML Internship — June – July 2025 (6 Weeks)
Edunet Foundation

- Trained and evaluated 4 supervised ML models (Logistic Regression, Random Forest, SVM, XGBoost), achieving 88% F1-score on real-world dataset with iterative feedback and GitHub version control

Certifications

- **Gen AI Academy Completion Certificate** — Hack2skill — *May 2025* | View Credential
- **Google Cloud Agentic AI Day** — Hack2skill — *Sep 2025* | View Credential
- **Code Veda** — Geek Room ADGIPS — *Sep 2025* | View Credential
- **TATA Crucible Campus Quiz 2025** — Unstop — *Nov 2025* | View Credential